

100 NLP Command Prompts for RL Preference Data

Reinforcement-Learning Volume Rendering Transfer Function — generated and parsed against the real commands.py pipeline (parse_command + apply_command), starting from default_params() each time. 50 direct-grammar prompts (parsed with parser="rule") and 50 natural-language prompts (parsed with parser="llm", model qwen2.5:7b, falling back to the rule parser on failure per the existing code path).

Part 1 — Direct grammar prompts (rule parser)

1. “increase opacity for bone slightly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.66

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'slightly'}`

2. “increase opacity for bone moderately”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.74

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'moderately'}`

3. “increase opacity for bone strongly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}`

4. “decrease opacity for bone slightly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.51

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'slightly'}`

5. “increase opacity for fat strongly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.62

`parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}`

6. “decrease opacity for fat moderately”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.03

`parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'moderately'}`

7. “increase opacity for spongy slightly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.40

`parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'slightly'}`

8. “decrease opacity for soft tissue strongly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): opacity 0.15 -> 0.06

`parsed: {'target': 'soft', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'strongly'}`

9. “high opacity bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.85

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}`

10. “low opacity for bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.15

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}`

11. “medium opacity spongy”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.50

`parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'set', 'level': 'medium'}`

12. “high opacity fat”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.85

`parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}`

13. “low opacity for soft tissue”

Parser: rule **Status:** ok **Parameters triggered:** (no measurable change) `parsed: {'target': 'soft', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}`

14. “medium opacity for fat”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.50

`parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'set', 'level': 'medium'}`

15. “high opacity for spongy”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.85

`parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}`

16. “low opacity air”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): center -100 -> -1000 HU; width 60 -> 80; opacity 0.05 -> 0.15

`parsed: {'target': 'air', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}`

17. “show only bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.00
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.70

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'show_only', 'strength': None}`

18. “show only spongy”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.00
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.70
- peak 3 (900 HU): opacity 0.60 -> 0.00

`parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'show_only', 'strength': None}`

19. “show only soft tissue”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.00
- peak 1 (40 HU): opacity 0.15 -> 0.70
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.00

```
parsed: {'target': 'soft', 'attribute': 'opacity', 'direction': 'show_only', 'strength': None}
```

20. “show only fat”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.70
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.00

```
parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'show_only', 'strength': None}
```

21. “show only bone and spongy”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.00
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.70
- peak 3 (900 HU): opacity 0.60 -> 0.70

```
parsed: {'target': ['bone', 'spongy'], 'attribute': 'opacity', 'direction': 'show_only', 'strength': None}
```

22. “show only fat and soft tissue”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.70
- peak 1 (40 HU): opacity 0.15 -> 0.70
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.00

```
parsed: {'target': ['fat', 'soft'], 'attribute': 'opacity', 'direction': 'show_only', 'strength': None}
```

23. “high opacity spongy, low opacity bones”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.85
- peak 3 (900 HU): opacity 0.60 -> 0.15

```
parsed: {'compound': [{'target': 'spongy', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}, {'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}]}
```

24. “low opacity for fat, high opacity for bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.15
- peak 3 (900 HU): opacity 0.60 -> 0.85

```
parsed: {'compound': [{'target': 'fat', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}, {'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}]}
```

25. “medium opacity soft, low opacity bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): opacity 0.15 -> 0.50
- peak 3 (900 HU): opacity 0.60 -> 0.15

```
parsed: {'compound': [{'target': 'soft', 'attribute': 'opacity', 'direction': 'set', 'level': 'medium'}, {'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}]}
```

26. “high opacity bone, low opacity fat, medium opacity spongy”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.15
- peak 2 (300 HU): opacity 0.30 -> 0.50
- peak 3 (900 HU): opacity 0.60 -> 0.85

```
parsed: {'compound': [{'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}, {'target': 'fat', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}, {'target': 'spongy', 'attribute': 'opacity', 'direction': 'set', 'level': 'medium'}]}
```

27. “sharpen the bone peak”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): width 280 -> 186

```
parsed: {'target': 'bone', 'attribute': 'width', 'direction': 'decrease', 'strength': 'moderately'}
```

28. “soften soft tissue”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): width 40 -> 166

```
parsed: {'target': 'soft', 'attribute': 'width', 'direction': 'increase', 'strength': 'moderately'}
```

29. “increase width for fat”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): width 60 -> 179

```
parsed: {'target': 'fat', 'attribute': 'width', 'direction': 'increase', 'strength': 'moderately'}
```

30. “sharpen the spongy peak strongly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): width 80 -> 38

```
parsed: {'target': 'spongy', 'attribute': 'width', 'direction': 'decrease', 'strength': 'strongly'}
```

31. “soften bone slightly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): width 280 -> 298

```
parsed: {'target': 'bone', 'attribute': 'width', 'direction': 'increase', 'strength': 'slightly'}
```

32. “increase width for bone moderately”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): width 280 -> 322

```
parsed: {'target': 'bone', 'attribute': 'width', 'direction': 'increase', 'strength': 'moderately'}
```

33. “brighten bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): rgb (0.95,0.95,0.90) -> (0.97,0.97,0.94)

```
parsed: {'target': 'bone', 'attribute': 'brightness', 'direction': 'increase', 'strength': 'moderately'}
```

34. “darken fat”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): rgb (0.95,0.90,0.60) -> (0.62,0.58,0.39)

```
parsed: {'target': 'fat', 'attribute': 'brightness', 'direction': 'decrease', 'strength': 'moderately'}
```

35. “low brightness for spongy”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): rgb (0.90,0.80,0.60) -> (0.15,0.15,0.15)

```
parsed: {'target': 'spongy', 'attribute': 'brightness', 'direction': 'set', 'level': 'low'}
```

36. “brighten soft tissue strongly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): rgb (0.85,0.35,0.35) -> (0.94,0.74,0.74)

`parsed: {'target': 'soft', 'attribute': 'brightness', 'direction': 'increase', 'strength': 'strongly'}`

37. “darken bone slightly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): rgb (0.95,0.95,0.90) -> (0.81,0.81,0.77)

`parsed: {'target': 'bone', 'attribute': 'brightness', 'direction': 'decrease', 'strength': 'slightly'}`

38. “high brightness for fat”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): rgb (0.95,0.90,0.60) -> (0.85,0.85,0.85)

`parsed: {'target': 'fat', 'attribute': 'brightness', 'direction': 'set', 'level': 'high'}`

39. “shift bone’s center up”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): center 900 -> 1285 HU

`parsed: {'target': 'bone', 'attribute': 'center', 'direction': 'increase', 'strength': 'moderately'}`

40. “move fat down”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): center -100 -> -432 HU

`parsed: {'target': 'fat', 'attribute': 'center', 'direction': 'decrease', 'strength': 'moderately'}`

41. “shift spongy’s position down”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): center 300 -> 170 HU

`parsed: {'target': 'spongy', 'attribute': 'center', 'direction': 'decrease', 'strength': 'moderately'}`

42. “move soft tissue’s center up strongly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): center 40 -> 170 HU

`parsed: {'target': 'soft', 'attribute': 'center', 'direction': 'increase', 'strength': 'moderately'}`

43. “shift bone’s center down slightly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): center 900 -> 600 HU

`parsed: {'target': 'bone', 'attribute': 'center', 'direction': 'decrease', 'strength': 'moderately'}`

44. “move fat’s position up moderately”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): center -100 -> -30 HU

`parsed: {'target': 'fat', 'attribute': 'center', 'direction': 'increase', 'strength': 'moderately'}`

45. “increase opacity for spongy strongly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.72

`parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}`

46. “decrease opacity for bone moderately”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.39

```
parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'moderately'}
```

47. “show only fat and bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.70
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.70

```
parsed: {'target': ['fat', 'bone'], 'attribute': 'opacity', 'direction': 'show_only', 'strength': None}
```

48. “low opacity for fat, medium opacity for spongy, high opacity for bone”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.15
- peak 2 (300 HU): opacity 0.30 -> 0.50
- peak 3 (900 HU): opacity 0.60 -> 0.85

```
parsed: {'compound': [{'target': 'fat', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}, {'target': 'spongy', 'attribute': 'opacity', 'direction': 'set', 'level': 'medium'}, {'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}]}
```

49. “increase width for spongy slightly”

Parser: rule **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): width 80 -> 128

```
parsed: {'target': 'spongy', 'attribute': 'width', 'direction': 'increase', 'strength': 'slightly'}
```

50. “reset”

Parser: rule **Status:** ok **Detail:** reset to default_params() **parsed:** {'target': None, 'attribute': None, 'direction': 'reset', 'strength': None}

Part 2 — Natural phrasing prompts (LLM parser)

1. “make the bones really stand out”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

```
parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}
```

2. “I want to see mostly skeleton”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

```
parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}
```

3. “can you fade out the fat a bit”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.04

```
parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'slightly'}
```

4. “hide the soft tissue almost completely”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): opacity 0.15 -> 0.06

```
parsed: {'target': 'soft', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'strongly'}
```

5. “turn up how visible the bone is”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.74

```
parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'moderately'}
```

6. “tone down the fat”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.03

`parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'moderately'}`

7. “let’s really boost the bone visibility”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}`

8. “just show me the bones”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.00
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.70

`parsed: {'target': 'bone', 'attribute': None, 'direction': 'show_only', 'strength': None}`

9. “only show soft tissue”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: ‘only show soft tissue’

10. “hide everything except bone and spongy”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.00
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.70
- peak 3 (900 HU): opacity 0.60 -> 0.70

`parsed: {'target': ['bone', 'spongy'], 'attribute': None, 'direction': 'show_only', 'strength': None}`

11. “make it look like an x-ray of just the skeleton”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}`

12. “I want high visibility on the spongy bone and low on the hard bone”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.85
- peak 3 (900 HU): opacity 0.60 -> 0.15

`parsed: {'compound': [{'target': 'spongy', 'attribute': 'opacity', 'direction': 'set', 'level': 'high'}, {'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}]}`

13. “crank up bone and dial down fat”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}`

14. “give me a good balance, medium bone and low fat”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.15
- peak 3 (900 HU): opacity 0.60 -> 0.50

`parsed: {'compound': [{'target': 'bone', 'attribute': 'opacity', 'direction': 'set', 'level': 'medium'}, {'target': 'fat', 'attribute': 'opacity', 'direction': 'set', 'level': 'low'}]}`

15. “make the bone edges crisper”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): width 280 -> 240

`parsed: {'target': 'bone', 'attribute': 'width', 'direction': 'decrease', 'strength': 'slightly'}`

16. “blur the soft tissue a little”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): width 40 -> 36

`parsed: {'target': 'soft', 'attribute': 'width', 'direction': 'decrease', 'strength': 'slightly'}`

17. “make the fat region blend in more”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.03

`parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'moderately'}`

18. “sharpen up the spongy bone strongly”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): width 80 -> 38

`parsed: {'target': 'spongy', 'attribute': 'width', 'direction': 'decrease', 'strength': 'strongly'}`

19. “soften the bone boundary slightly”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): width 280 -> 240

`parsed: {'target': 'bone', 'attribute': 'width', 'direction': 'decrease', 'strength': 'slightly'}`

20. “widen the fat band”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: ‘widen the fat band’

21. “make the bones brighter”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): rgb (0.95,0.95,0.90) -> (0.97,0.97,0.94)

`parsed: {'target': 'bone', 'attribute': 'brightness', 'direction': 'increase', 'strength': 'moderately'}`

22. “make the fat darker”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): rgb (0.95,0.90,0.60) -> (0.81,0.77,0.51)

`parsed: {'target': 'fat', 'attribute': 'brightness', 'direction': 'decrease', 'strength': 'slightly'}`

23. “dim the spongy bone”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.20

`parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'moderately'}`

24. “really light up the soft tissue”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): rgb (0.85,0.35,0.35) -> (0.94,0.74,0.74)

`parsed: {'target': 'soft', 'attribute': 'brightness', 'direction': 'increase', 'strength': 'strongly'}`

25. “darken the bone a touch”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): rgb (0.95,0.95,0.90) -> (0.81,0.81,0.77)

`parsed: {'target': 'bone', 'attribute': 'brightness', 'direction': 'decrease', 'strength': 'slightly'}`

26. “make fat glow more”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): rgb (0.95,0.90,0.60) -> (0.97,0.94,0.74)

`parsed: {'target': 'fat', 'attribute': 'brightness', 'direction': 'increase', 'strength': 'moderately'}`

27. “nudge the bone’s position higher”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: "nudge the bone’s position higher"

28. “move the fat’s center lower”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): center -100 -> -432 HU

`parsed: {'target': 'fat', 'attribute': 'center', 'direction': 'decrease', 'strength': 'moderately'}`

29. “shift the spongy tissue down a bit”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): center 300 -> 170 HU

`parsed: {'target': 'spongy', 'attribute': 'center', 'direction': 'decrease', 'strength': 'slightly'}`

30. “push the soft tissue’s center way up”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): center 40 -> 170 HU

`parsed: {'target': 'soft', 'attribute': 'center', 'direction': 'increase', 'strength': 'strongly'}`

31. “bring the bone center down slightly”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): center 900 -> 608 HU

`parsed: {'target': 'bone', 'attribute': 'center', 'direction': 'decrease', 'strength': 'slightly'}`

32. “move fat’s position upward moderately”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): center -100 -> -30 HU

`parsed: {'target': 'fat', 'attribute': 'center', 'direction': 'increase', 'strength': 'moderately'}`

33. “make the fat almost invisible”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.02

`parsed: {'target': 'fat', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'strongly'}`

34. “I want a lot more bone showing”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}`

35. “blend the fat and soft tissue together”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 1 (40 HU): opacity 0.15 -> 0.10

`parsed: {'target': 'soft', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'moderately'}`

36. “make the spongy bone barely visible”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.26

`parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'slightly'}`

37. “start over”

Parser: llm **Status:** ok **Detail:** reset to default_params() parsed: {'direction': 'reset', 'target': None, 'attribute': None, 'strength': None}

38. “reset everything back to normal”

Parser: llm **Status:** ok **Detail:** reset to default_params() parsed: {'target': None, 'attribute': None, 'direction': 'reset', 'strength': None}

39. “make it pretty”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 2 (300 HU): opacity 0.30 -> 0.54

parsed: {'target': 'spongy', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'moderately'}

40. “surprise me”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: ‘surprise me’

41. “make the render look nicer”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: ‘make the render look nicer’

42. “can you improve the image quality”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: ‘can you improve the image quality’

43. “show me just the fatty tissue”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.70
- peak 1 (40 HU): opacity 0.15 -> 0.00
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.00

parsed: {'target': 'fat', 'attribute': None, 'direction': 'show_only', 'strength': None}

44. “I only want to see soft tissue and fat”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): opacity 0.05 -> 0.70
- peak 1 (40 HU): opacity 0.15 -> 0.70
- peak 2 (300 HU): opacity 0.30 -> 0.00
- peak 3 (900 HU): opacity 0.60 -> 0.00

parsed: {'target': ['soft', 'fat'], 'attribute': None, 'direction': 'show_only', 'strength': None}

45. “increase how thick the bone band is”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): width 280 -> 322

parsed: {'target': 'bone', 'attribute': 'width', 'direction': 'increase', 'strength': 'moderately'}

46. “make the skeleton pop more strongly”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.84

parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'increase', 'strength': 'strongly'}

47. “fade the air out completely”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 0 (-100 HU): center -100 -> -1000 HU; width 60 -> 80; opacity 0.05 -> 0.02

parsed: {'target': 'air', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'strongly'}

48. “boost bone opacity to the max”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: ‘boost bone opacity to the max’

49. “let’s go back to the default view”

Parser: llm **Status:** PARSE FAILED **Detail:** rule parser cannot parse: "let’s go back to the default view"

50. “make the bone almost see-through”

Parser: llm **Status:** ok **Parameters triggered:**

- peak 3 (900 HU): opacity 0.60 -> 0.51

`parsed: {'target': 'bone', 'attribute': 'opacity', 'direction': 'decrease', 'strength': 'slightly'}`